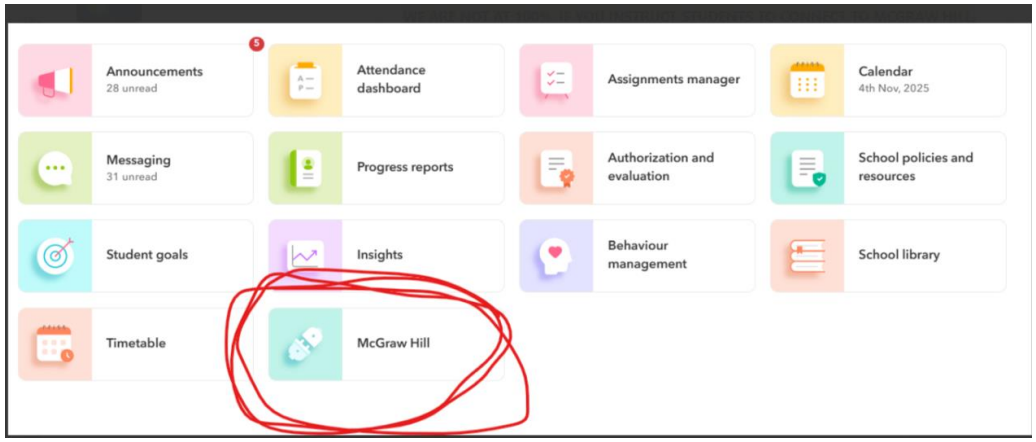
Week 11: 9th – 13th November 2026

Subject: Mathematics Accelerated **Unit Title: Unit 1:** Numeric reasoning, equations and geometric relationships **Grade: 7**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Summative Differentiation	Resources and links for Parents
Lesson 1: Understand similarity	I can identify and explain similar figures by comparing their corresponding angles and side lengths, and use proportional relationships to determine whether two shapes are similar.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	**Practice links/review links:** **Lesson 1: Understand similarity** **Lesson 2: Angle angle similarity**
Lesson 2: Angle angle similarity	How can translations, rotations, and reflections transform a shape while preserving its size and shape, and how can we describe these transformations mathematically		
Lesson 3 and 4: GRASP task	I can apply my mathematical knowledge of integer and rational-number operations, equations and inequalities, angle relationships, and geometric transformations to design a playable escape-room challenge with logical clues, a geometric lock and a final code. I can communicate my mathematical reasoning, verify solutions and use AI responsibly to support and improve my design.		
Teacher's notes/ Reminders	- Make use of the notebook and workbook provided. - PowerPoints and additional notes/practice will be uploaded on teams – under the shared/files section. - Please submit the GRASP after the two lessons. - **Homework: Complete ALL Worksheets based on Transformations. Your teacher will share the worksheets on teams.** **Mc Graw hill: Can be accessed on Toddle:**		



Week 11

Technology Integration دمج التكنولوجيا	Teams Class Textbook/workbook.
Cross-curricular links لروابط بين المواد الدراسية	Design/Digital Design: puzzle and lock-mechanism design connects to systems thinking and user experience.
External Assessment Integration دمج التقييمات الخارجية	MAP – Common core 6+
Wellbeing/ ATL Skills ة النفسية/ أساليب التعلم	Thinking skills: Identify structure and equivalence when solving equations and analysing transformations; recognize patterns in integer/rational-number operations. Communication skills: Use precise algebraic and geometric vocabulary and notation. Self-management skills: Work carefully and check solutions when solving multi-step equations and inequalities. Social skills: Collaborate to critique and justify whether two figures are congruent or similar.
Standards المعايير	8.G.A.4, 8.G.A.5

Inquiry/ TOK Question البحث والاستقصاء أسئلة	How can translations, rotations, and reflections transform a shape while preserving its size and shape, and how can we describe these transformations mathematically?
Success Criteria (Measurable) معايير النجاح	**Lesson 1:** 🟦 All students must recognize corresponding sides and corresponding angles. 🔵 Most students should explain how similar figures have the same shape but may have different sizes. 🔺 Some students could justify why two figures are similar using corresponding angles and proportional sides. **Lesson 2:** 🟦 All students must state what it means for two triangles to be similar. 🔵 Most students should use the AA Similarity Theorem to determine whether two triangles are similar. 🔺 Some students could apply AA similarity to multi-step geometric problems and real-life situations. **Lesson 3 and 4:** 🟦 All students must create a mathematics escape-room theme and explain the mission. 🔵 Most students should use an AI tool to generate or improve a puzzle idea and check that its mathematical content is accurate. 🔺 Some students could justify why I selected, adapted or rejected an AI-generated suggestion and reflect on how mathematical reasoning and AI contributed to the final product.
National Identity Domain مجالات الربط بالهوية الوطنية	Volunteering التطوع Respect الاحترام History التاريخ
	المجالArea: students explore examples of logic puzzles, Islamic geometric patterns, and UAE STEM/innovation challenges as inspiration for well-designed problems.
	التطبيقImplementation: Suggested AI Tools for Student Use (Research and Innovation): AI chatbots for clue-wording and prompt practice (e.g., school-approved AI assistants); AI-powered puzzle/logic generators; teacher-curated AI tool comparison activity (students select, use, evaluate, and justify their AI tool choice as part of the GRASPS reflection). Note: All AI tools must be used in line with the ENS Academic Integrity Policy. Students must document all AI tool usage, prompts, and responses in their puzzle portfolio.